Capital Bikeshare - From Raw Data to A/B Test Decision

Members and casual riders use the bikeshare system in fundamentally different ways

Portfolio: serhiy-dranko-portfolio.carrd.co

GitHub: github.com/serhiy-dranko/capital-bikeshare-pipeline

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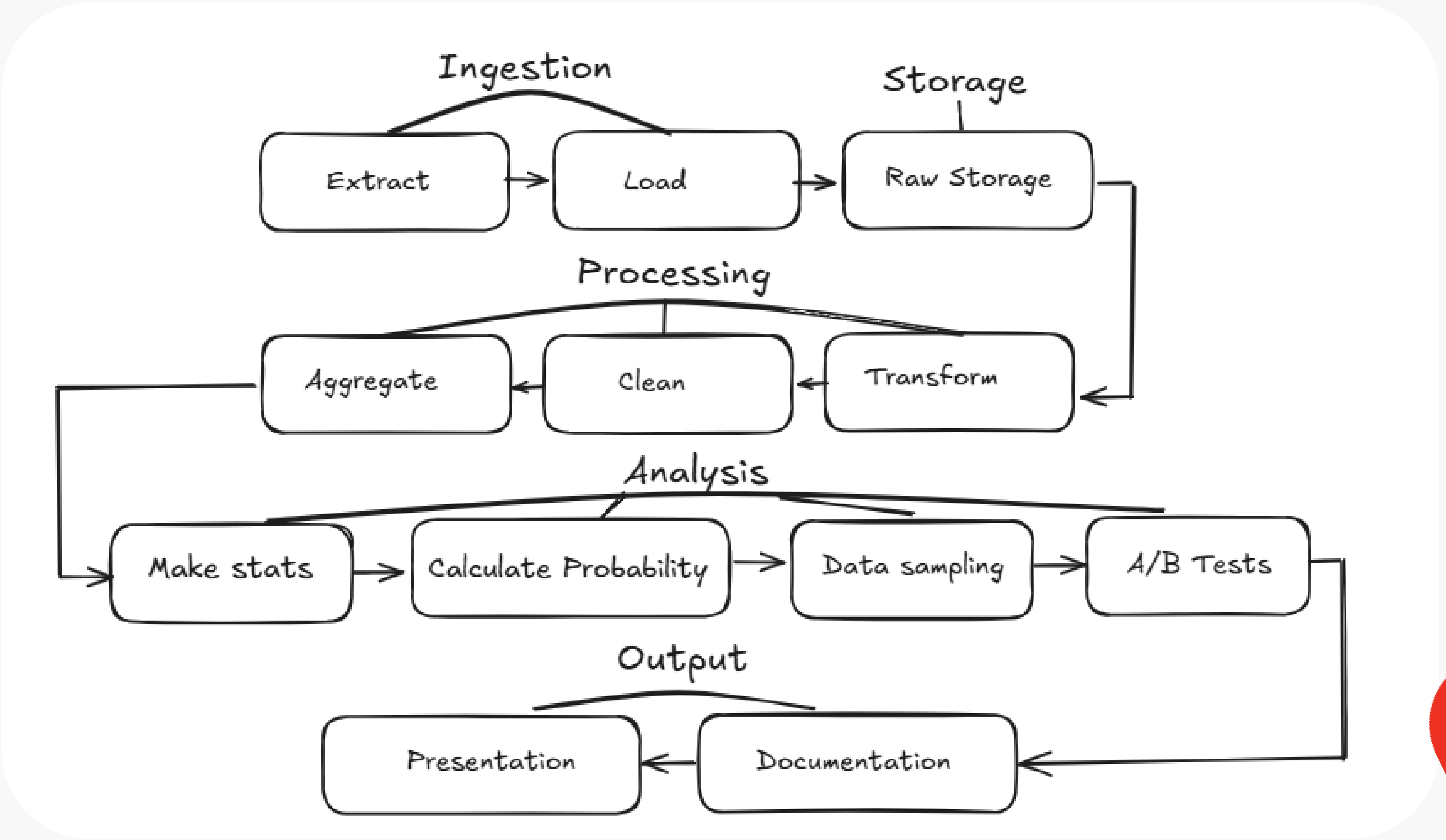
The core question: Are members and casual riders using this system for the same thing? The answer is clearly no, that gap is the business opportunity.

This matters because the system is already show us two very different use cases under one structure. The window between commute peaks, middle is where casual riders concentrate but members largely disappear, which is the clearest signal for dynamic pricing experimentation.

Source: Capital Bikeshare, Sep 2023 · n = 97044 rows · Imported at 12.06.2026 09:47:34

Capital Bikeshare pipeline

Capital Bikeshare - trip history September 2023 publicly available at capitalbikeshare.com



Capital Bikeshare - trip history September 2023 have 97044 valid trips (97.04% of the data)

*during limitation Google sheets took only 100K rows

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Capital Bikeshare Data Quality

Total raw rows ingested: 100,000

Rows excluded: 2955 rows.

A trip was marked valid if it had a duration between 1 and 180 minutes,
a non-blank ride ID, and a non-blank rider type. Anything outside those rules was flagged FALSE
and excluded from all downstream analysis.

667 trips are missing end_lat / end_lng and not many of them are docked bikes.
Missing coordinates on a docked bike make sense but classics bikes did.
The trip ended somewhere unknown, which means destination analysis is silently incomplete.

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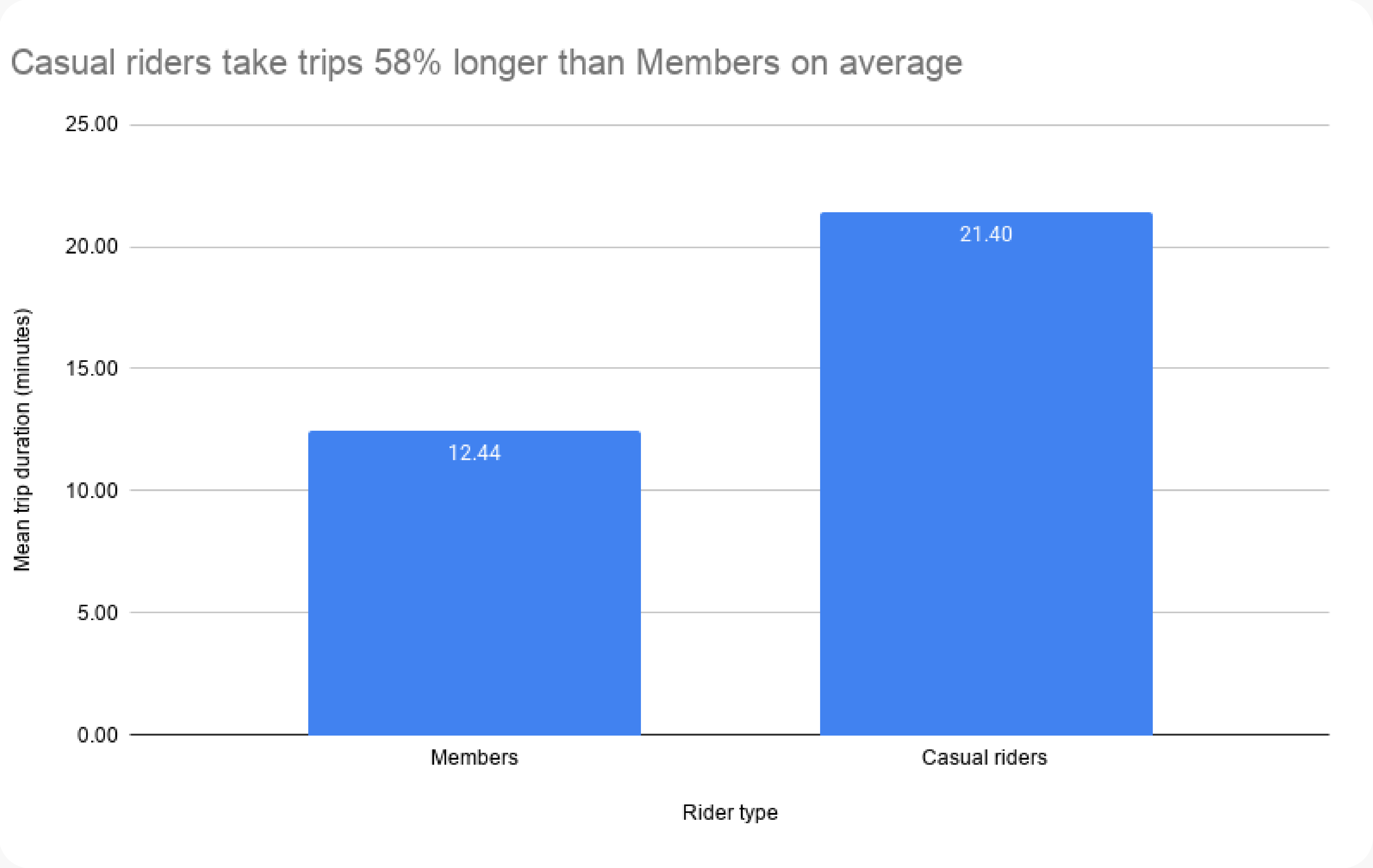
Casual riders take trips twice as long as members



Casual riders average 21.4 minutes per trip versus 12.4 minutes for Members.

This suggests the system serves two fundamentally different populations - commuters who need short, reliable rides and leisure riders who are exploring the city.

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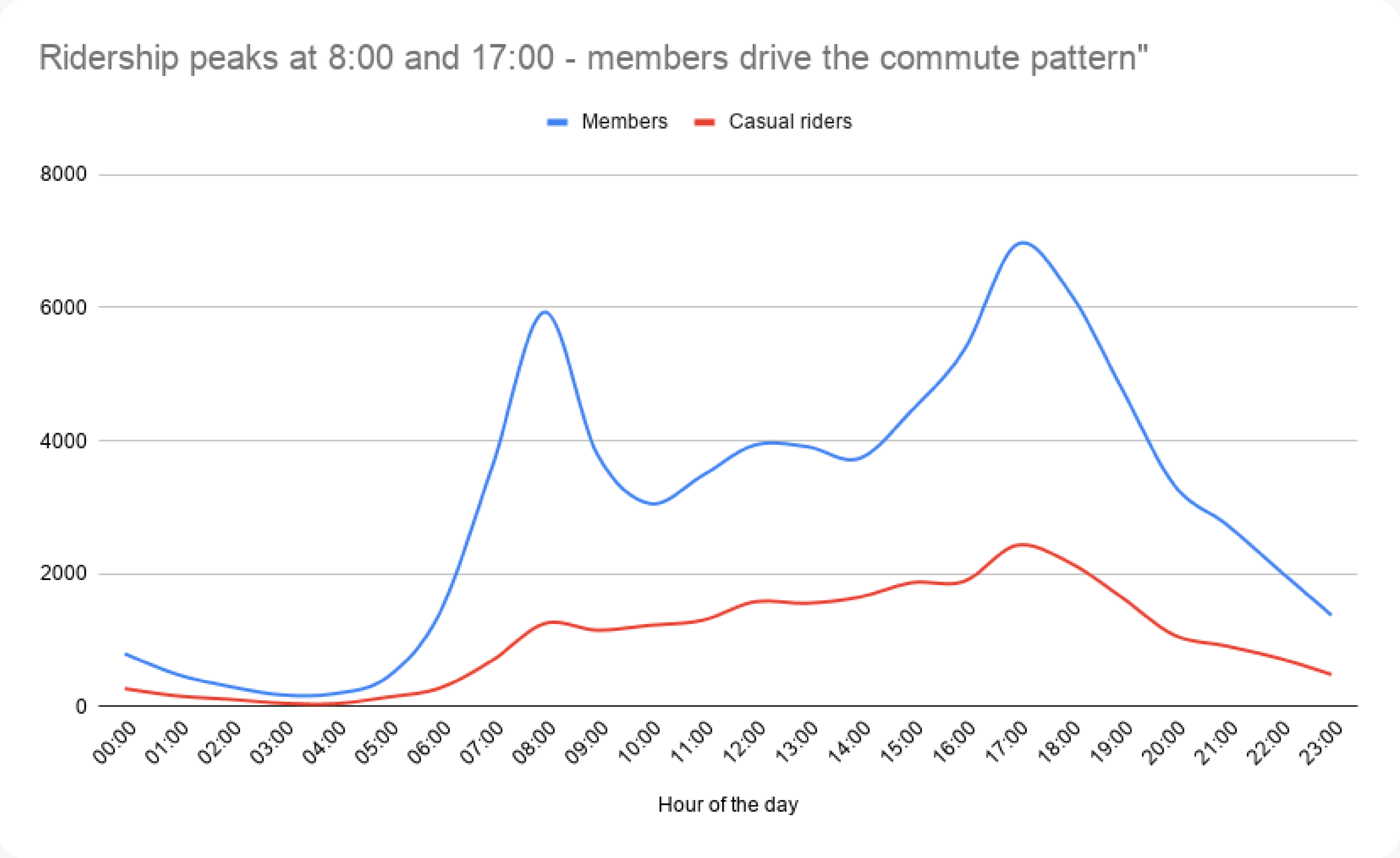
Bikeshare becomes a commuter tool after 8:00 - then goes quiet until 17:00 at working days



Trip volume spikes at 8:00 and 17:00, following the pattern of a typical office commute during the working days.

In time between this two peak hours we can see lower quantity of usage by Casual rider so we can ajust our price policy.

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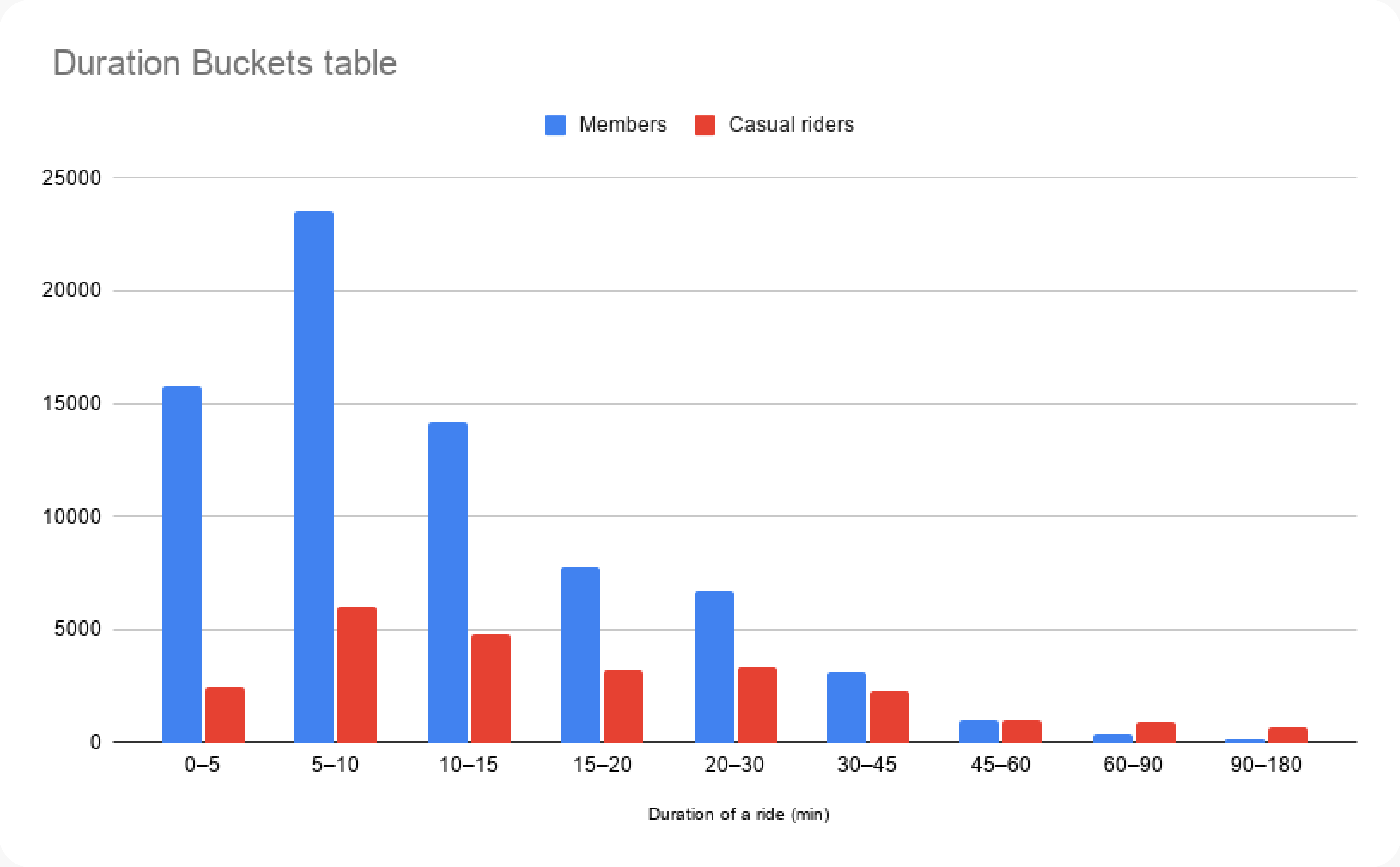
The gap between Members and Casual riders narrows as trip duration increases.



Most member trips fall within the 5-15 minute range, while casual riders are more likely to take longer trips 15-45 minute range.

This suggests that members primarily use bikes for commuting and short transportation needs, while casual riders use them for leisure, indicating opportunities to convert casual riders into memberships through targeted offers for frequent or longer duration users.

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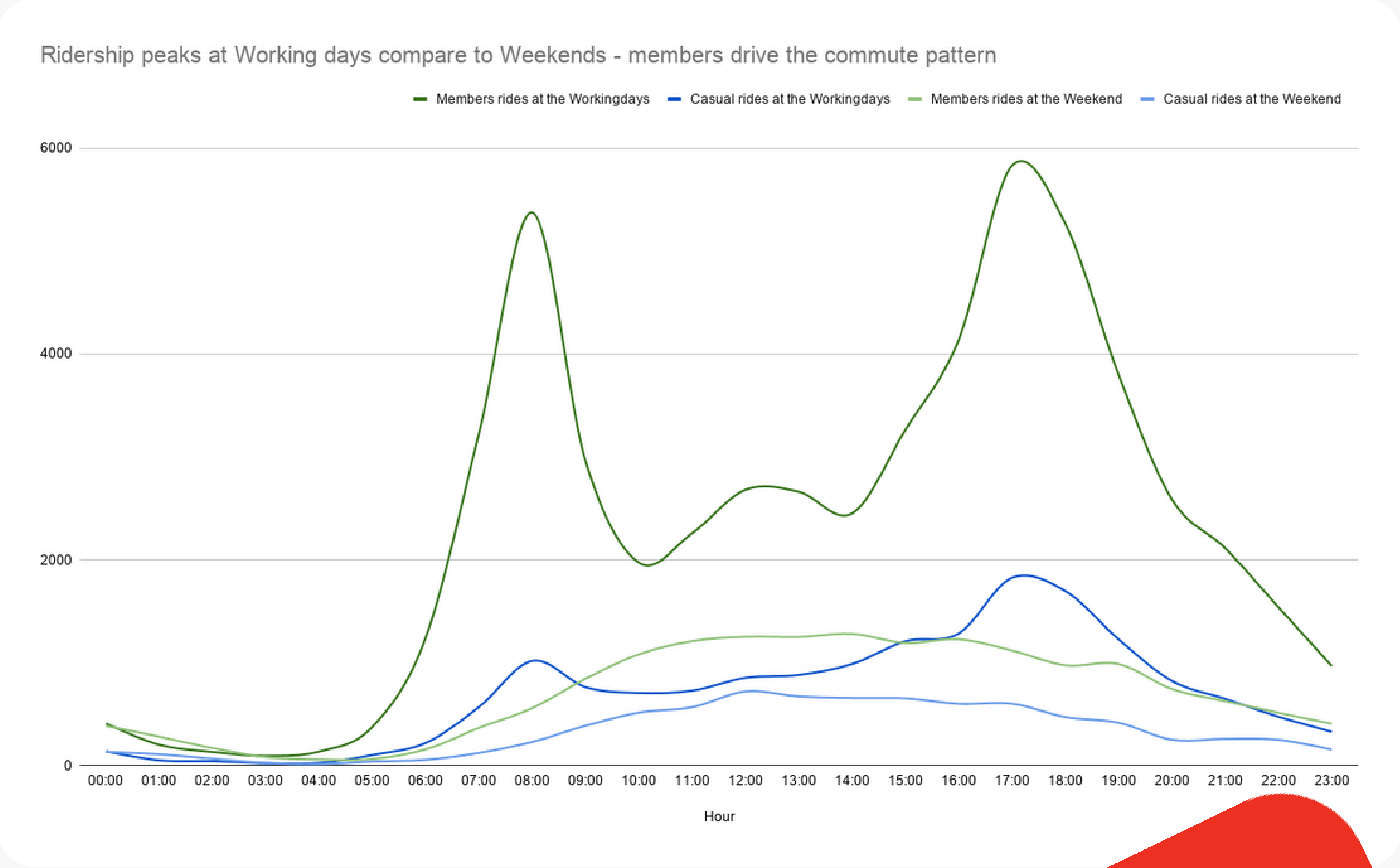
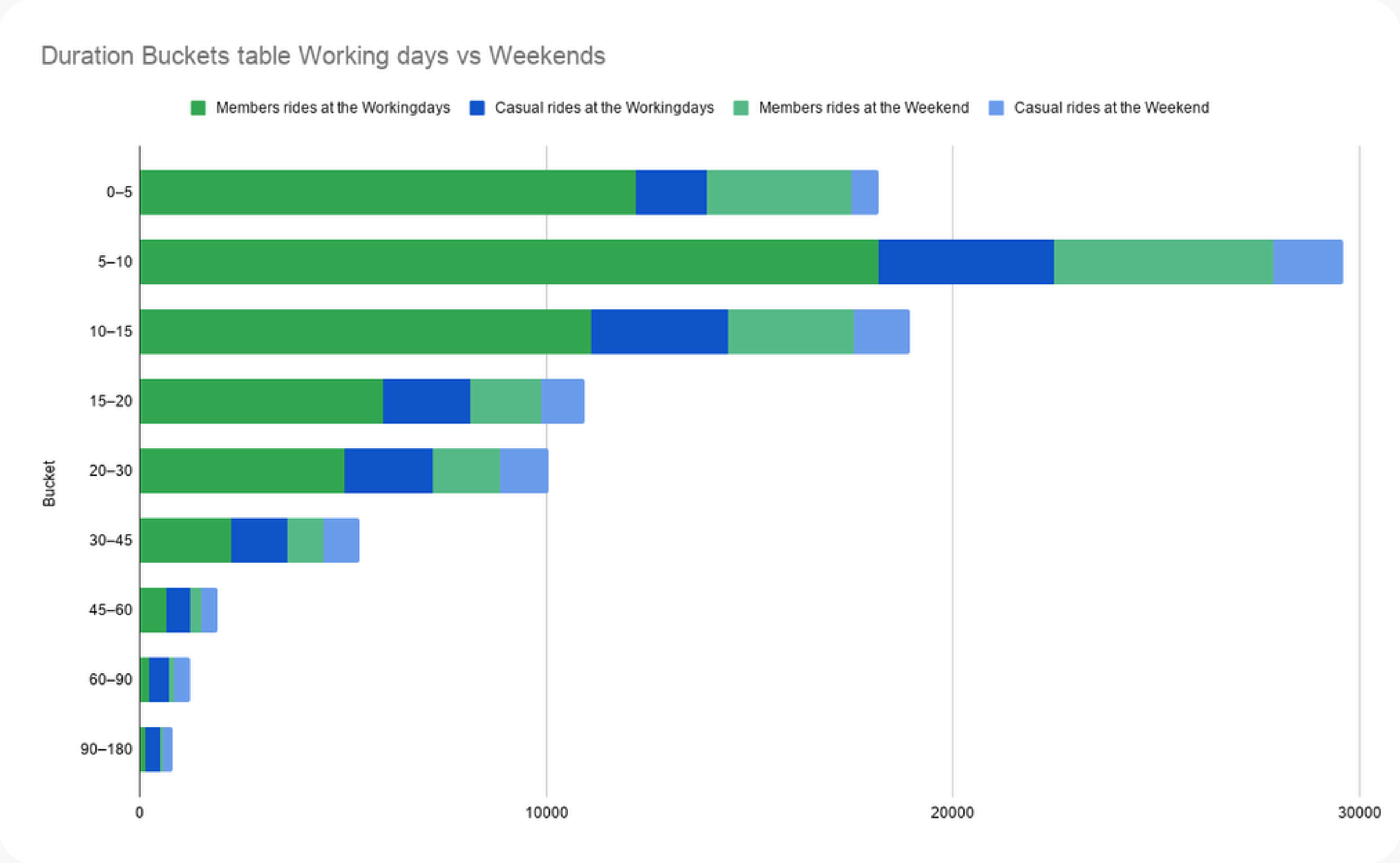
The gap between Members & Casual commuters in terms of part of the week.

Trip volume spikes at 8:00 and 17:00, following the pattern of a typical office commute during the working days. If we look more deeper into the data we can see that on the Weekends, our peak hours is between 12:00 and 14:00.

But we don't see much difference between casuals and members compared to the Working days data. This means that, casuals can be converted to members with targeted offers for work days short rides between 8:00 and 17:00.





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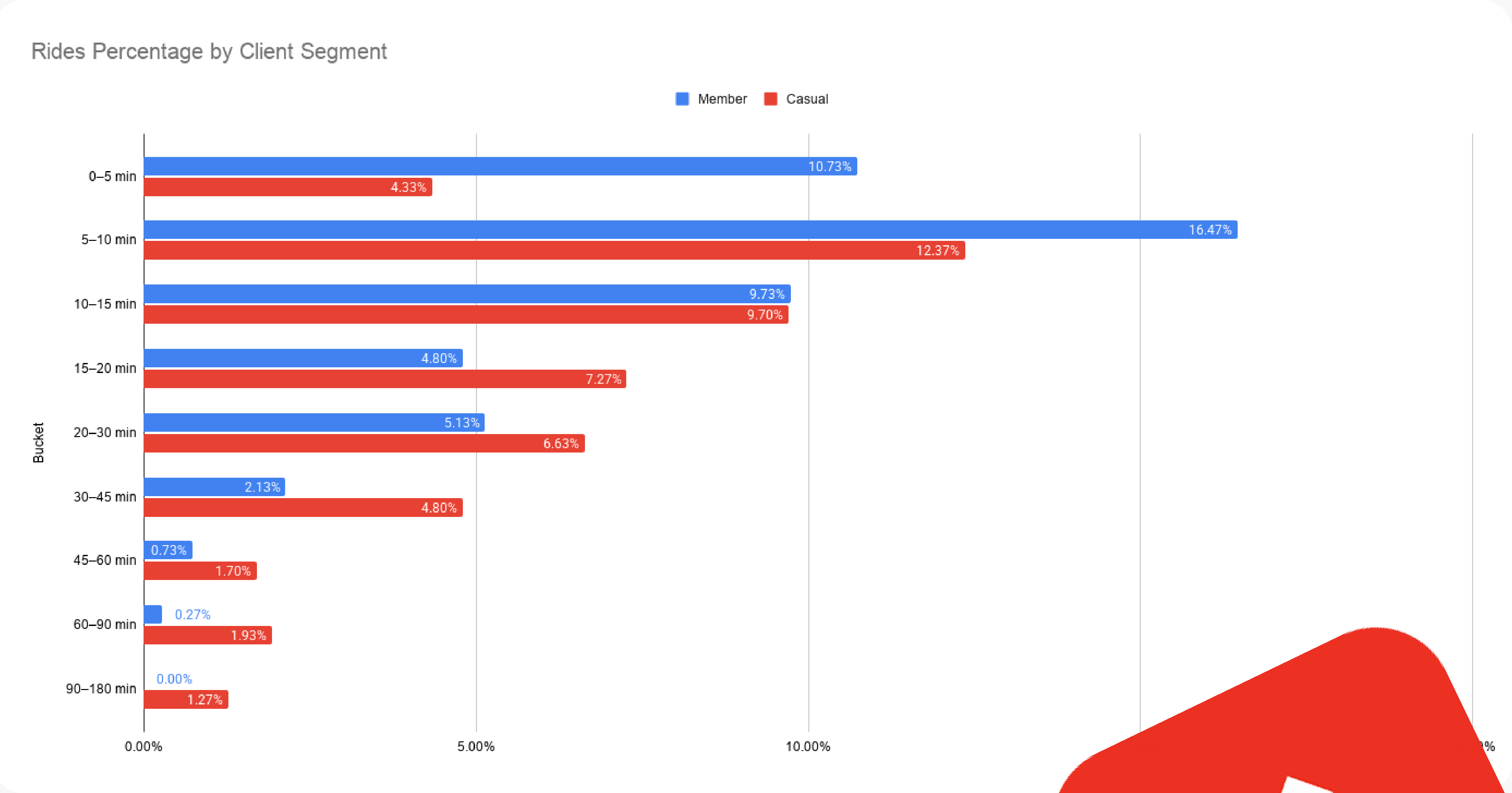
Same Bike, Different Purpose Members Ride 74% Shorter.

The effect is real but not dramatic: a Cohen's d of 0.51 means the two groups overlap substantially, and the confound check confirms bike type doesn't explain the gap (classic: 12.3 vs 22.3 min; electric: 12.5 vs 17.7 min the member-casual split persists across both).

The 12.4% of casual users already riding under 15 minutes are the easiest conversion target for membership growth.

Classic Bike	Members	Casual riders
Mean duration	12.37	21.52
Std deviation	10.57	22.85
Sample size	1,500	1,500
Difference	9.16 (min)	
p-value	0.0000	
Cohen's d	0.51	
Verdict	Significant	



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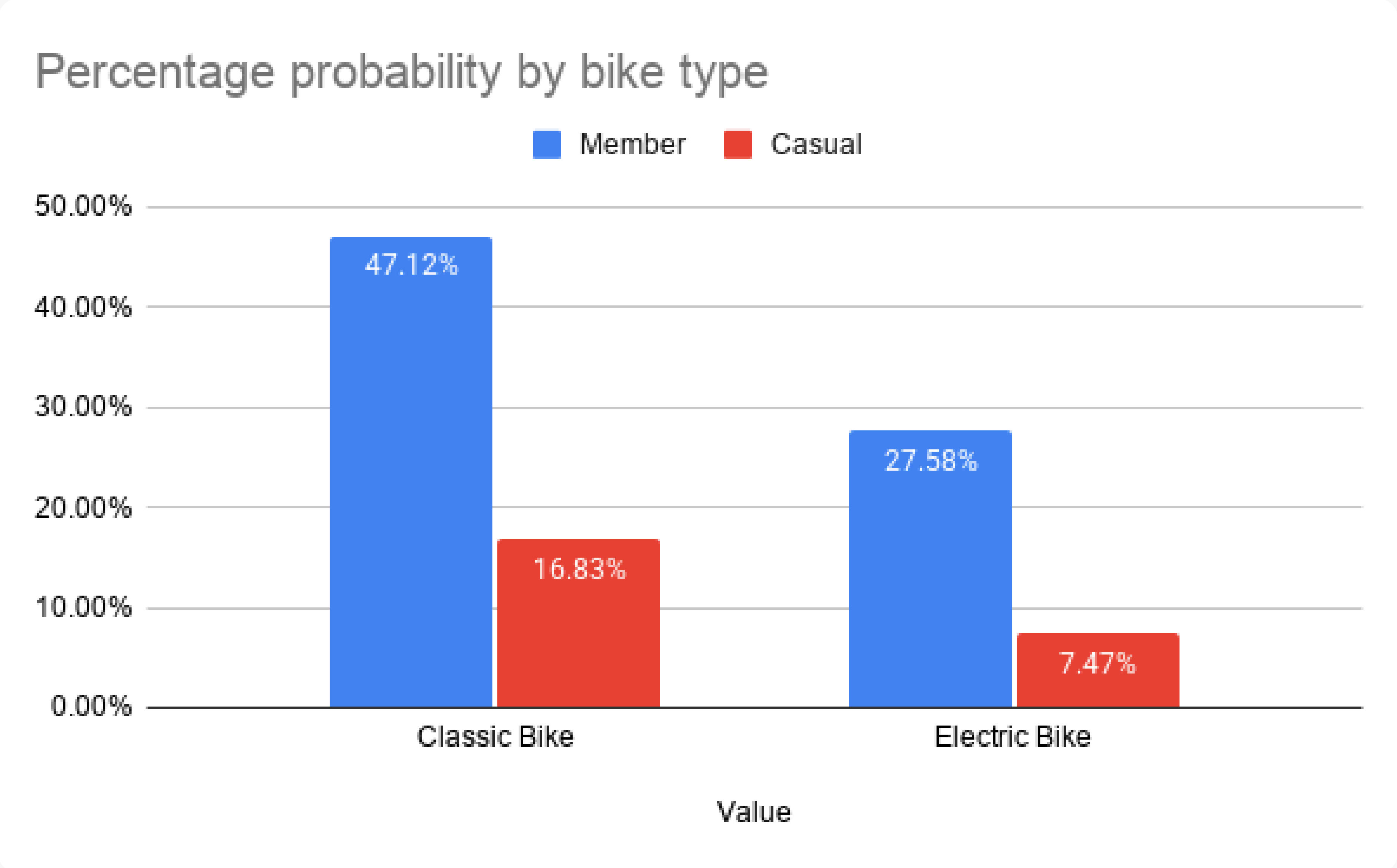
Classic bikes show high probability regardless of customer type



	Classic Bike	Electric Bike	Docked bike	Row Total
Member	47.12%	27.58%	0.00%	**74.70%**
Casual	16.83%	7.47%	1.00%	**25.30%**
Column Total	**63.95%**	**35.05%**	**1.00%**	**100.00%**

Members for 74.7% of all rides, with classic bikes being the dominant choice (47.1% of Total trips), while Casual riders represent 25.3% of rides and use classic bikes most frequently as well (16.8% of Total trips).

This indicates that membership users drive the majority of bike usage, so a business decision maker could focus on converting casual riders into members by promoting benefits that match frequent riding behavior

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GitHub and Portfolio

Everything I've presented today is publicly documented and reproducible. Anyone can open my GitHub repo, download the source data from Capital Bikeshare, and rebuild this pipeline from scratch following the README and pipeline log.

Link: github.com/serhiy-dranko/capital-bikeshare-pipeline

Also portfolio page is itself part of the project. It documents a working data pipeline built on Capital Bikeshare data. Below you'll also find a summary of findings and plans for future projects.

Link: serhiy-dranko-portfolio.carrd.co

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What I learned during the project

The most unexpected finding was how clearly structured documentation reduced the barrier to understanding for a non-technical audience. Good documentation, it turns out, is not just an internal tool. It is the kind a presentation.

Data cleaning and filter management in Google Sheets consumed a disproportionate share of the project timeline. What appeared to be straightforward validation. Removing blank rows, flagging duration outliers, isolating valid records in fact required constant manual coordination across tabs.

What I would do differently.

1. Replace formulas with pivot tables and Measures.
2. Switch to other programm (like Excel). Elswhere dataset fits without sampling.
3. Visualise in Power BI tool. Interactive dashboards connected directly to the data source.

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What I'd Investigate Next

September definitely has a good weather, casual numbers are likely inflated by tourists who may disappear when weather become colder.

If the duration gap narrows in the winter because only commuters are left in data.
That completely changes the conversion strategy. So we should definitely analyze more data before making any decisions.

Analysing a full year of Capital Bikeshare data would mean working with well over one million rows.
Which this project already confirmed Google Sheets cannot handle.
So the immediate skill gap is moving the pipeline into a tool built for that scale: Excel for a first step, SQL or Python for anything beyond it.
The pipeline logic I built across these project transfers directly to other programs.
Where it may change in the environment and runs in and the volume of data it needs to hold

Thank you!

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Pipeline: docs.google.com/spreadsheets/d/1GTRWs7EQf-0B6a5CGs0x70nKAOVCuKq1gbzI7UB1h5s/edit?usp=sharing